Project Title: Typing Accuracy Checker

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GitHub Repository: [sajidali108/typing-accuracy-checker-8086: 8086 Assembly-based typing accuracy checker that compares user input with predefined text, counts mistakes, calculates correctness, and displays performance results.]

Objective

The objective of this project is to develop a Typing Accuracy Checker using 8086 Assembly Language.

This program compares the user's typed input with a predefined sentence and calculates typing accuracy by counting correct characters and mistakes.

This project demonstrates practical implementation of assembly language concepts such as:

- Input and output handling
- String comparison
- Loops
- Conditional jumps
- Character processing

The project helps in understanding low-level programming and computer organization concepts.

Tools Used

1. NASM

Used to write and compile assembly language code.

2. DOSBox

Used to execute assembly programs in DOS environment.

3. AFD Debugger

Used to debug and analyze program execution.

4. GitHub

Used for project hosting and submission.

5. 8086 Assembly Language

Core programming language used for project development.

Compilation Command

nasm typing.asm -o typing.com

Execution Command

typing.com

Code Explanation

The project is developed using 8086 Assembly Language.

The code is divided into different sections:

1. org 100h

This tells the assembler that the program is a `.COM` file and execution starts from memory location 100h.

2. Data Declaration

In this section, messages, predefined sentence, input storage, and counters are declared.

Examples:

- Display messages
- Input array
- Correct character counter
- Mistake counter

3. Display Instructions

The program uses DOS interrupt `INT 21h` with function `09h` to display text on screen.

Purpose:

- Show instructions to user
- Display results

4. Input Handling

The program takes user input character-by-character using DOS interrupt `INT 21h`.

Purpose:

- Store typed characters
- Process user input

5. Comparison Logic

The entered input is compared with the predefined sentence using `CMP` instruction.

If characters match:

- Correct counter increases

If characters do not match:

- Mistake counter increases

6. Looping

Loops are used to check all characters one by one.

This ensures complete comparison of input.

7. Result Evaluation

The program checks total mistakes and displays result as:

- Perfect Typing
- Good
- Poor

using conditional jump instructions.

8. Program Termination

The program ends using DOS interrupt:

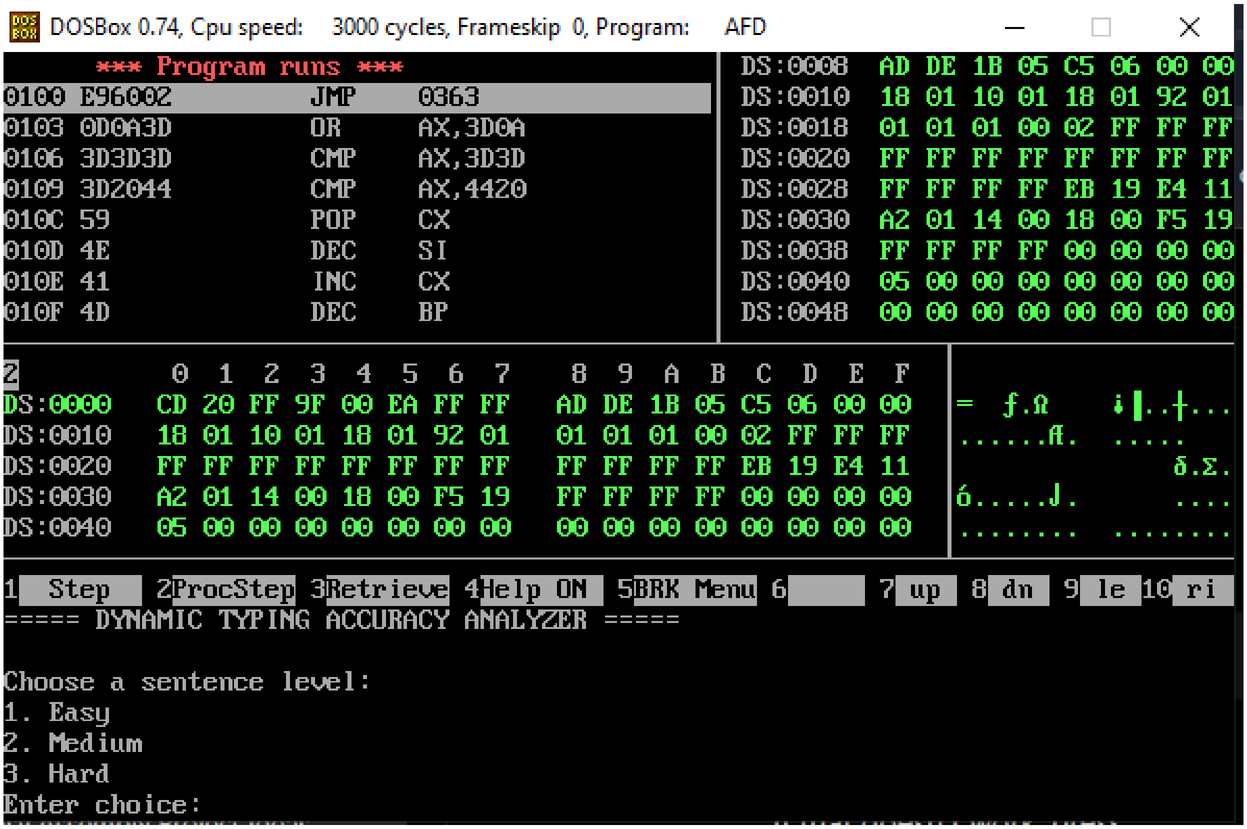
mov ah,4ch
int 21h

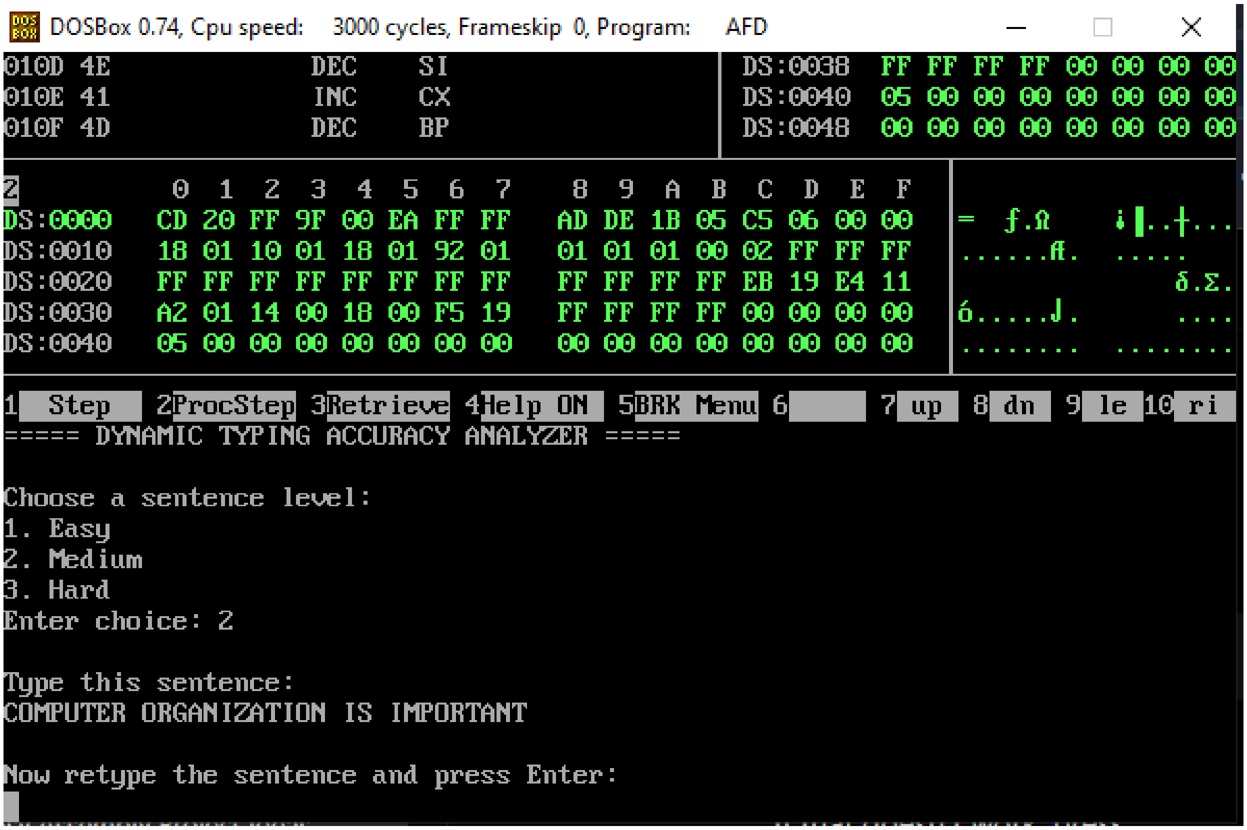
This safely exits the program.

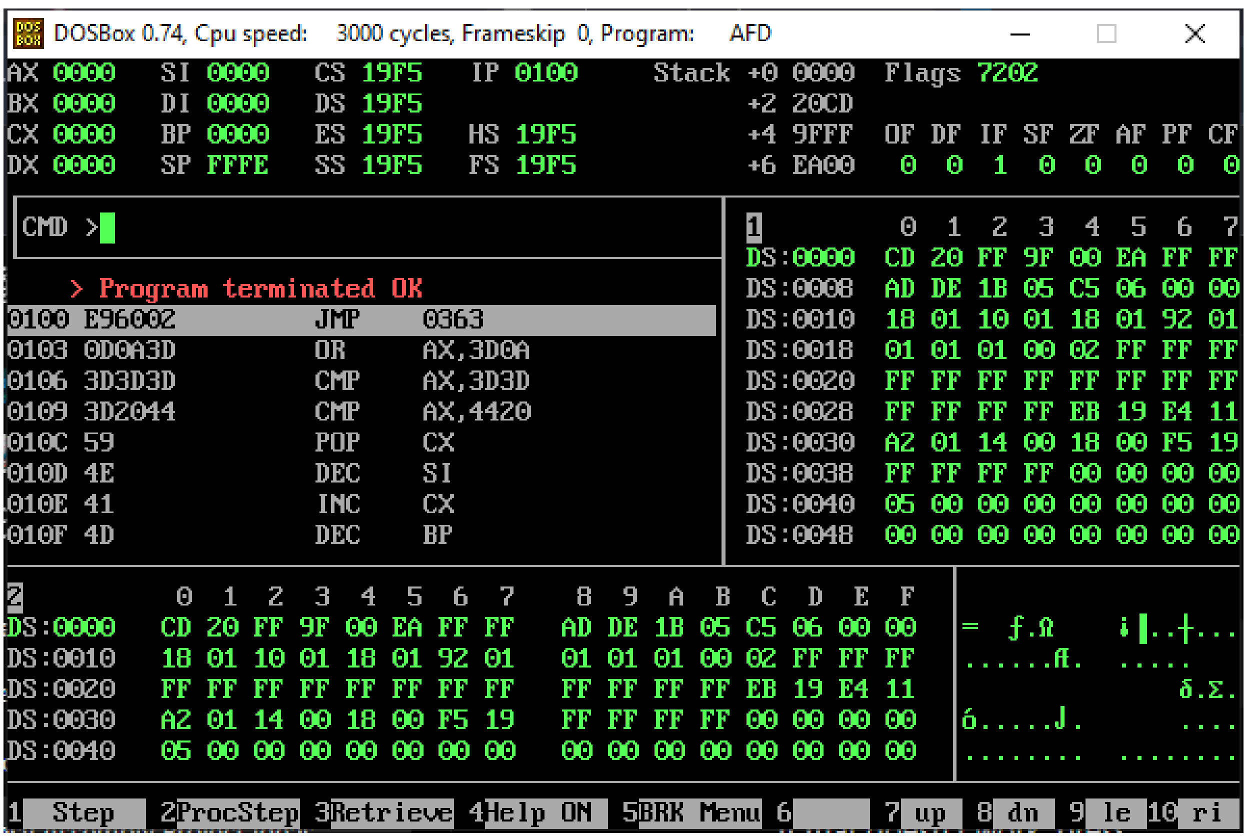
Screenshots

```
C:\>nasm typing.asm -o typing.com

C:\>SS
```

```
C:\>nasm typing.asm -o typing.com

C:\>typing

===== DYNAMIC TYPING ACCURACY ANALYZER =====

Choose a sentence level:
1. Easy
2. Medium
3. Hard
Enter choice: 1

Type this sentence:
ASSEMBLY IS EASY

Now retype the sentence and press Enter:
ASSEmbly is EASY

===== RESULT =====
Expected Characters: 16
Typed Characters: 16
Correct Characters: 10
Mistakes: 6
Final Result: Needs Practice
C:\>s_
```


DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: AFD
*** Program runs ***
0100 E96002 JMP 0363
0103 0D0A3D OR AX,3D0A
0106 3D3D3D CMP AX,3D3D
0109 3D2044 CMP AX,4420
010C 59 POP CX
010D 4E DEC SI
010E 41 INC CX
010F 4D DEC BP
DS:0008 AD DE 1B 05 C5 06 00 00
DS:0010 18 01 10 01 18 01 92 01
DS:0018 01 01 01 00 02 FF FF FF
DS:0020 FF FF FF FF FF FF FF FF
DS:0028 FF FF FF FF EB 19 E4 11
DS:0030 A2 01 14 00 18 00 F5 19
DS:0038 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00
DS:0048 00 00 00 00 00 00 00 00
DS:0000 CD 20 FF 9F 00 EA FF FF AD DE 1B 05 C5 06 00 00
DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF
DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 E4 11
DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
1 Step 2 ProcStep 3 Retrieve 4 Help ON 5 BRK Menu 6 7 up 8 dn 9 le 10 ri
===== DYNAMIC TYPING ACCURACY ANALYZER =====
Choose a sentence level:
1. Easy
2. Medium
3. Hard
Enter choice:


DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: AFD
010D 4E DEC SI
010E 41 INC CX
010F 4D DEC BP
DS:0038 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00
DS:0048 00 00 00 00 00 00 00 00
2 0 1 2 3 4 5 6 7 8 9 A B C D E F
DS:0000 CD 20 FF 9F 00 EA FF FF AD DE 1B 05 C5 06 00 00
DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF
DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 E4 11
DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri
===== DYNAMIC TYPING ACCURACY ANALYZER =====
Choose a sentence level:
1. Easy
2. Medium
3. Hard
Enter choice: 2
Type this sentence:
COMPUTER ORGANIZATION IS IMPORTANT
Now retype the sentence and press Enter:


DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: AFD
AX 0000 SI 0000 CS 19F5 IP 0100 Stack +0 0000 Flags 7202
BX 0000 DI 0000 DS 19F5 +2 20CD
CX 0000 BP 0000 ES 19F5 HS 19F5 +4 9FFF OF DF IF SF ZF AF PF CF
DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 0 0 0 0
CMD >
> Program terminated OK
0100 E96002 JMP 0363
0103 0D0A3D OR AX,3D0A
0106 3D3D3D CMP AX,3D3D
0109 3D2044 CMP AX,4420
010C 59 POP CX
010D 4E DEC SI
010E 41 INC CX
010F 4D DEC BP
1 0 1 2 3 4 5 6 7
DS:0000 CD 20 FF 9F 00 EA FF FF
DS:0008 AD DE 1B 05 C5 06 00 00
DS:0010 18 01 10 01 18 01 92 01
DS:0018 01 01 01 00 02 FF FF FF
DS:0020 FF FF FF FF FF FF FF FF
DS:0028 FF FF FF FF EB 19 E4 11
DS:0030 A2 01 14 00 18 00 F5 19
DS:0038 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00
DS:0048 00 00 00 00 00 00 00 00
2 0 1 2 3 4 5 6 7 8 9 A B C D E F
DS:0000 CD 20 FF 9F 00 EA FF FF AD DE 1B 05 C5 06 00 00
DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF
DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 E4 11
DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00
DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6 7 up 8 dn 9 le 10 ri